CS23E01 / EMBEDDED SYSTEMS AND INTERNET OF THINGS

- Session: July-Nov 2026 (B.E. CSE – VII sem)

Activities for Lab-6:

Topics: timer events, serial communication, connecting to display in **Embedded C**.

Software/simulator required: MCU 8051 IDE with SDCC compiler or equivalent.

// For basic Embedded 'C' programs refer to the pdf/material shared.

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1. Demonstrate the working of serial communication (transmit and receive characters using UART)

2. Implement timer based delay using Timer for accurate delays for toggling an LED at a port.

3. Imagine a switch is connected to P3 while an LED is connected to P1. Alternatively switch on and off the LED as the switch is pressed again and again.

4. Interface with an LCD/7-segment display

5. using 7-segment LCD display (virtual hardware) with delay for displaying numbers from 0 to 9 in a loop.

6. implement a simple digital counter that counts the number of times a button (high on P1) is pressed and displays the count on any display panel).

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